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SCALE depletion capabilities for molten salt reactors and other liquid-fueled systems☆

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ABSTRACT

Nuclear reactor systems that use fuel dissolved in a liquid have the potential for enhanced safety characteristics, improved fuel-cycle outcomes, and more efficient isotope-production configurations. In these reactor systems, the fueled liquid may simultaneously undergo irradiation, physical and chemical removal processes, and fueling. The modeling and simulation of this transmutation and decay with material additions and removals is an ongoing research area. An accurate simulation tool is critical to the reactor and fuel-cycle design, reactor deployment, and source-term characterization for these advanced reactor systems. The work described herein involved implementing, testing, and applying the capability to perform reactor physics simulations within the Oak Ridge National Laboratory-developed SCALE suite for nuclear systems analyses and design, leveraging much of its pedigree in quality-assurance and reactor-analysis capabilities. The functionalities to simulate irradiation with material feeds and removals had been added in ORIGEN, and the TRITON reactor physics sequence was extended to calculate the total removed material and track external nonirradiated mixtures to estimate separate processing or waste streams. Results from these capabilities align with analytical expectations obtained from ORIGEN for simplified test cases and with expectations for a molten salt reactor application. This implementation, available with the SCALE 6.3 release, provides for a more efficient and accurate material accountability methodology, allowing for the characterization, design, and analysis of the complete isotopic material inventory of advanced liquid-fueled systems for a variety of applications.

1. Introduction

Liquid-fueled molten salt reactors (MSRs) are among several potential advanced nuclear reactor technologies currently being proposed by reactor developers. Favored for their potential for inherent safety characteristics and improved fuel-cycle outcomes, MSR designs range in power level (microreactor up to 1 GWth), carrier salt type (e.g., chloride or fluoride salt), fuel type (e.g., uranium or thorium), neutron energy spectrum (e.g., thermal, intermediate, and fast), and fuel cycle. This flexibility provides for varied deployment, operation, and fueling/refueling scenarios that may generate additional gaseous and solid waste streams unique to MSRs and other technologies containing a liquid fuel form. Predicting the material flowing through these types of systems is important for reactor design, safety analyses, source term characterization, safeguards analysis, and processing system design (e.g., off-gas system [OGS] design) (Betzler et al., 2019b).

Some liquid-fueled MSR designs will be loaded with the majority of their total fuel inventory at reactor startup. For these systems,

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[1] Work completed while employed at Oak Ridge National Laboratory.

[2] A typical light-water reactor has discrete fuel assemblies that move into and out of the reactor at given intervals, whereas liquid fuel systems may contain the majority or entirety of this fuel material at any given time.

[3] Short-time phenomena are more important for characterizing spatial dependencies of fuel compositions during steady-state operation or for characterizing dynamic events.

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the fuel-cycle characteristics—the additional fuel and waste streams moving into and out of the liquid fuel mixture—are key parameters to accurately capture in the reactor physics simulation.[2] Appropriately accounting for the flow of the fuel and waste streams will provide for a more accurate characterization of reactor physics parameters over long time periods (i.e., months to years) as fission products and actinides accumulate in the liquid fuel mixture. These material flows are not to be confused with the physical movement of the liquid fuel into and out of irradiation zones (e.g., from the core to heat-exchange machinery), which presents a separate characterization problem on timescales of seconds to hours. However, resolving these short-time phenomena is beyond the scope of this paper.[3]

No established tool currently exists for reactor physics and fuel-cycle design and evaluation of liquid-fueled MSRs, but significant works in fast and thermal MSR design and analyses have yielded workable tools that perform relevant reactor physics analyses (Bauman et al., 1971; Nuttin et al., 2005; Sheu et al., 2013; Powers et al., 2013; Fiorina et al., 2013; Aufiero et al., 2013; Heuer et al., 2014; Ahmad et al., 2015; Park et al., 2015; Jeong et al., 2016; Rykhlevskii et al., 2019a; Hombourger et al., 2020; Zhuang et al., 2020). In general, these tools use or extend the capabilities of typical transmutation and decay solvers used for the analysis of solid-fueled reactor systems in one of two ways: (1) using the existing solvers with additional bookkeeping scripts[4] to effectively model semicontinuous operations using iterative simulations and (2) extending the existing solvers to include continuous removal and feed of isotopes or elements during irradiation (Aufiero et al., 2013). This includes the development of the generic Python script known as ChemTriton (Betzler et al., 2017c), which provides a flexible framework to simulate a variety of MSR fuel-cycle concepts (Betzler et al., 2016b,a, 2017d; Gentry et al., 2017; Skirpan et al., 2017; Betzler et al., 2017b, 2018) by iteratively using SCALE (Wieselquist et al., 2020) to characterize the material flow behavior.

The work described herein involved integrating the capabilities demonstrated by ChemTriton directly into SCALE by extending existing solvers to quantify and track continuous feeds and fractional removals during the irradiation of a given mixture. This steady integration process consisted of methodically moving through precursor drift analyses (Betzler et al., 2017a), various testing versions of SCALE and beta releases (Betzler et al., 2019a), and external collaborator analysis activities (Rykhlevskii et al., 2019b; Vicente Valdez et al., 2020; Soares et al., 2020; Cornejo et al., 2021; Lo et al., 2022). The underlying quality assurance pedigree, reactor system analysis capabilities, and SCALE user base provide a robust basis on which to build a reliable liquid-fueled reactor analysis capability. SCALE's depletion and decay solver ORIGEN (Gauld et al., 2011) and the reactor physics sequence TRITON (DeHart and Bowman, 2011) are industry standards for source term characterization and lattice physics analyses. TRITON performs coupled transport-depletion simulations with a well-validated approach (Ilas et al., 2022): using any of SCALE's neutron transport kernels for the neutron transport calculations in which neutron flux and cross sections are determined, and using ORIGEN to calculate nuclide concentration changes during irradiation or decay. ORIGEN had been extended to simulate irradiation with continuous feeds and removals characteristic of liquid-fueled systems (Isotalo and Wieselquist, 2015). In TRITON, the data exchange and computation flow to simulate the liquid-fueled reactors molten salt reactor with fractional nuclide removal and continuous feed are identical for computing other solid-fueled reactors. However, inputs were added to define the flow of materials into and out of irradiated mixtures, and the tool was extended to track and decay quantities of removed materials. This two-step method requires recasting a removal rate as a feed rate to an external mixture representative of a storage tank, processing system, or waste form. This type of approach, coupled with the scaled flux method to account for the fuel's depletion rate, can predict isotopic compositions during irradiation to inform the follow-on system and fuel-cycle design decisions for advanced reactors (Bae et al., 2021; Betzler, 2021). Implementing these tools into SCALE stands to benefit a larger number of analysis tools that use ORIGEN for depletion calculations while also providing a basis to apply additional tools for optimization, sensitivity studies, safeguards analyses, and more resolved reactor physics analyses.

This article describes and demonstrates the SCALE 6.3 analysis capability for liquid-fueled systems. It also discusses the theoretical underpinnings of depletion with continuous removals (Section 2), the numerical implementations of these developed models (Section 3), the testing of the implementations (Section 3), and the simulation of representative MSR scenarios (Section 4). Comparisons to the Serpent approach (Leppänen et al., 2015) are provided (Section 4).

2. Theory

ORIGEN solves the system of ordinary differential equations that describes nuclide transmutation, depletion, and decay (Gauld et al., 2011),

$$\frac{dN_i}{dt} = \sum_{i'} \left(\lambda_{i'\to i} + \sigma_{i'\to i}\phi\right) N_{i'}(t) - \left(\lambda_i + \lambda_{i,\mathrm{rem}} + \sigma_i\phi\right) N_i(t) + S_{i,\mathrm{ext}}(t), \quad (1)$$

where

- N_i is the amount of nuclide i (atoms),
- $\lambda_{i'\to i}$ is the transition coefficient for the decay of nuclide i' resulting in nuclide i (1/s),
- $\sigma_{i'\to i}$ is the transition coefficient for the spectrum-averaged neutron-induced transmutation of nuclide i' resulting in nuclide i (barn),
- ϕ is the angle- and energy-integrated time-dependent neutron flux (neutrons/cm^2 s),
- λ_i is the total decay constant for nuclide i (1/s),
- σ_i is the total energy-averaged loss coefficient for nuclide i in a neutron flux field (barn),
- $\lambda_{i,\mathrm{rem}}$ is the continuous removal constant for nuclide i (1/s), and
- $S_{i,\mathrm{ext}}$ is the time-dependent external feed term (atoms/s).

Note that ORIGEN has always had the capability to include both removal $\lambda_{i,\mathrm{rem}}$ and feed $S_{i,\mathrm{ext}}$ in a single-mixture problem. Described herein is the new capability introduced for modeling MSRs, including the capability to define a set of "flows" that can occur between arbitrary mixtures, implemented in the reactor physics sequence, TRITON, which manages the ORIGEN calculations for each mixture. Two types of flows are supported:

1. A removal from one mixture m' to another mixture m described by nuclide-dependent removal constants $\lambda_{i,\mathrm{rem}}^{m'\to m}$.
2. An external feed, $S_{i,\mathrm{ext}}^{m}(t)$, into mixture m.

Including these terms leads to the following set of equations for each mixture m and nuclide i.

$$\frac{dN_i^m}{dt} = \sum_{i'} \left(\lambda_{i'\to i} + \sigma_{i'\to i}\phi\right) N_{i'}(t) + \sum_{m'} \lambda_{i,\mathrm{rem}}^{m'\to m} N_i^{m'}(t) - \left(\lambda_i + \lambda_{i,\mathrm{rem}}^{m} + \sigma_i\phi\right) N_i(t) + S_{i,\mathrm{ext}}^{m}(t), \quad (2)$$

where $\lambda_{i,\mathrm{rem}}^{m} = \sum_{m'} \lambda_{i,\mathrm{rem}}^{m\to m'}$ is total removal constant for nuclide i in mixture m. The external feed flow for multiple mixtures is straightforward, simply connecting to the existing ORIGEN capability. The removal from one mixture to another is more complex. This connection could be

[4] Scripts vary in complexity and may track more complex chemical processes, reactor conditions, and physics dependencies.

resolved by creating a large transition matrix in which off-diagonal blocks correspond to the removal flows.

$$\begin{pmatrix} \frac{\vec{N}^1}{dt} \\ \frac{\vec{N}^2}{dt} \\ \vdots \\ \frac{\vec{N}^m}{dt} \\ \vdots \\ \frac{\vec{N}^M}{dt} \end{pmatrix} = \begin{pmatrix} \mathbf{A}^1 & \mathbf{\Lambda}_{\text{rem}}^{2\to 1} & \cdots & \mathbf{\Lambda}_{\text{rem}}^{m\to 1} & \cdots & \mathbf{\Lambda}_{\text{rem}}^{M\to 1} \\ \mathbf{\Lambda}_{\text{rem}}^{1\to 2} & \mathbf{A}^2 & \cdots & \mathbf{\Lambda}_{\text{rem}}^{m\to 2} & \cdots & \mathbf{\Lambda}_{\text{rem}}^{M\to 2} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ \mathbf{\Lambda}_{\text{rem}}^{1\to m} & \mathbf{\Lambda}_{\text{rem}}^{2\to m} & \cdots & \mathbf{A}^m & \cdots & \mathbf{\Lambda}_{\text{rem}}^{M\to m} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{\Lambda}_{\text{rem}}^{1\to M} & \mathbf{\Lambda}_{\text{rem}}^{2\to M} & \cdots & \mathbf{\Lambda}_{\text{rem}}^{m\to M} & \cdots & \mathbf{A}^M \end{pmatrix} \begin{pmatrix} \vec{N}^1(t) \\ \vec{N}^2(t) \\ \vdots \\ \vec{N}^m(t) \\ \vdots \\ \vec{N}^M(t) \end{pmatrix} + \begin{pmatrix} \vec{S}_{\text{ext}}^1(t) \\ \vec{S}_{\text{ext}}^2(t) \\ \vdots \\ \vec{S}_{\text{ext}}^m(t) \\ \vdots \\ \vec{S}_{\text{ext}}^M(t) \end{pmatrix} \tag{3}$$

However, this calculation quickly grows intractable using either the legacy MATREX or modern CRAM ordinary differential equation solver because the system solve time increases superlinearly with respect to the system size. For this reason, this work employed an iterative solution in which the upper off-diagonal is lagged one iteration behind and recast as a feed term where s is an iteration index.

$$\frac{d\vec{N}_s^m}{dt} = \mathbf{A}^m \vec{N}_s^m(t) + \vec{S}_{i,\text{ext}}^m(t) + \vec{R}_{s-1}^m(t), \tag{4}$$

where the lagged removal source is given as

$$\vec{R}_{s-1}^m(t) = \sum_{m'} \mathbf{\Lambda}_{\text{rem},s-1}^{m'\to m} \vec{N}_{s-1}^{m'}(t).$$

Consider a dependency graph of mixtures. If there are no "cycles" (i.e., mixture A does not flow into mixture B, and simultaneously B flows back into A), then the dependency is a directed acyclic graph, which can be traversed in a specific order, and iterations are not needed. In terms of Eq. (3), it means the upper diagonal is zero. For cyclic dependency graphs (e.g., specific nuclides plate out from the fuel salt to cold surfaces but with decay), some daughter nuclides return from the surface to the fuel salt. This type of problem uses a stopping criteria based on the relative change in mass from one iteration to another falling below some threshold ϵ,

$$\left| \frac{x_s^m}{x_{s-1}^m} - 1 \right| \le \epsilon, \tag{5}$$

where x_s^m is the mixture mass at iteration s and the end of the calculation time step. Each time step is converged before moving to the next.

Two implementation aspects remain: (1) approximating removal rates for physical systems and (2) the methodology to numerically integrate those removal rates in the feed expression.

2.1. Approximating removal rates

The removal term supported in the current model implies an exponential-type behavior for the specific removal mechanism such that the following expression holds:

$$N_i(t_1) = N_i(t_0) e^{-\lambda_{i,\text{rem}}(t_1 - t_0)}. \tag{6}$$

Therefore, a trivial way to approximate a removal rate is with a known reduction in quantity over a specific time, *owing to this mechanism only*,

$$\lambda_{i,\text{rem}} = -\frac{\ln \frac{N_i(t_1)}{N_i(t_0)}}{t_1 - t_0}. \tag{7}$$

This type of removal is usually chemical or mechanical in nature and thus is typically defined by the *element*, not by nuclide. Thus it may be useful to imagine a stable nuclide that is transparent to neutrons when approximating this removal term. The removal term is then applied to all isotopes of the same element. However, in TRITON, the removal term can be applied to an isotope, an element, or all nuclides in the mixture.

In MSR reference literature (Hombourger et al., 2020), the removal is typically defined in terms of a cycle time for the flowing salt and an efficiency

$$\lambda_{\text{i,rem}}^{m'\to m} = -\frac{\ln\left(1 - \epsilon_i^{m'\to m}\right)}{T_{\text{cycle}}}, \tag{8}$$

where T_{cycle} is a cycle time. For removals from the main flow loop, the cycle time could be defined as the loop time; therefore, $\epsilon_i^{m'\to m}$ becomes the relative amount of nuclide i removed per pass from mixture m' and deposited in mixture m. For example, with an 11 s flow loop and 1 part per million atoms (ppm) of noble metals plating out per pass from the main fuel salt (mixture A) onto the heat exchanger's surface (mixture B),

$$\lambda_{\text{i,rem}}^{A\to B} = -\frac{\ln\left(1 - \epsilon_i^{A\to B}\right)}{T_{\text{cycle}}} = -\frac{\ln\left(1 - 10^{-6}\right)}{11} = 0.090909\ 1/\text{s},$$

This type of nuclide flow model implies that the probability of removal is inversely proportional to the cycle time, T_{cycle}. Thus for the heat exchanger example, reducing the flow speed by half would halve the rate of plate out. In reality, these physical mechanisms may be complex functions of local temperature, flow speed, and time-dependent chemical properties of the fuel salt (Vicente Valdez et al., 2020; Cornejo et al., 2021; Graham et al., 2020; Lee and Jessee, 2023). For this reason, users are encouraged to perform a rigorous uncertainty quantification based on variations in these types of removal approximations.

2.2. Approximating removal sources

ORIGEN's feed term has an important limitation: currently the MATREX and CRAM solvers support only a stepwise constant in time representation. This requirement is enforced when users specify external feed S_{ext}. However, for casting a removal (exponential in nature) to a feed (constant in nature), ORIGEN must provide an internal integration. A single nuclide's source from all other mixtures is given by

$$r_{s-1}^m(t) = \sum_{m'} \lambda_{\text{rem},s-1}^{m'\to m} N_{s-1}^{m'}(t). \tag{9}$$

ORIGEN must provide a single, stepwise constant average value as follows:

$$\bar{r}_{s-1}^m = \sum_{m'} \lambda_{\text{rem},s-1}^{m'\to m} \bar{N}_{s-1}^{m'}, \tag{10}$$

with

$$\bar{N}_{s-1}^{m'} = \frac{1}{t_1 - t_0} \int_{t_0}^{t_1} N_{s-1}^{m'}(t)\, dt$$

Currently, the only data ORIGEN has are the beginning and end of step values. Thus, trapezoid rule integration seems an obvious choice:

$$\bar{N}_{\text{lin}} = 0.5 N_0 + 0.5 N_1, \tag{11}$$

where $N_0 = N(t_0)$ and $N_1 = N(t_1)$. However, the underlying function in many cases is a quickly decaying exponential, and the trapezoid rule applied directly will poorly estimate the true behavior. In this extreme, a trapezoid rule integration applied to the log of the nuclide amount is a good approximation:

$$\ln \bar{N}_{\text{log}} = 0.5 \ln N_0 + 0.5 \ln N_1. \tag{12}$$

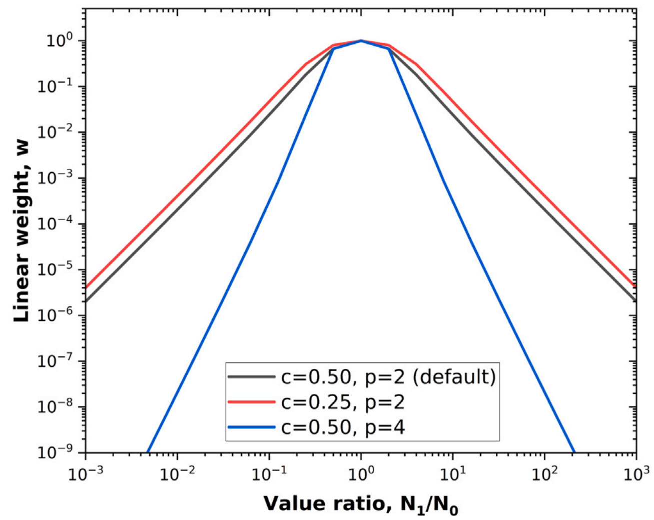

Fig. 1. Linear weight, w, as a function of N_1/N_0 ratio for various c and p values.

To achieve a smooth transition from linear to logarithmic integration formulas, a technique inspired by weighted essentially nonoscillatory techniques is used:

$$\bar{N} = w\bar{N}_{\text{lin}} + (1 - w)\bar{N}_{\text{log}}, \tag{13}$$

where the weight, w, is calculated as

$$w = \frac{1}{1 + c\left|1 - N_{\text{max}}/N_{\text{min}}\right|^p}, \tag{14}$$

where $N_{\text{max}} = \max(N_0, N_1)$, and $N_{\text{min}} = \min(N_0, N_1)$. By default, the method parameters are set to $c = 0.5$ and $p = 2$, controlling how fast the transition from linear to logarithmic occurs. Fig. 1 shows the behavior of the linear weight, w, which is 1.0 if the two values are identical and quickly decreases as the two values differ. For example, if the values differ by a factor of 10, then the linear weight is $w \approx 0.024$.

2.3. Caveats

The following subsections summarize some caveats to consider when using the current implementation. The removal of some of these caveats is planned for future versions of SCALE.

2.3.1. Volume handling

TRITON assumes that a model's geometry does not change over time. Although densities and compositions of materials can change, the volumes are constant at all times. If an external feed is applied in TRITON, then only the concentrations of the fed nuclides in the target material are modified. To maintain the criticality, molten salt reactors are designed to be refueled via the addition of fresh fuel salt. Some designs are equipped with a tank to hold the excess fuel salt. However, some reactors allow the fuel salt volume to increase over time. Because the increase in fuel volume cannot currently be directly modeled in TRITON, the volume increase must be indirectly considered by appropriately modifying the fuel concentrations and potentially the feed rate over time.

TRITON permits the specification of so-called *decay-only* mixtures, which allow the collection of removed nuclides and the decay of these outside the reactor. An example application case is the modeling of the OGS as a decay-only mixture: fission products are continuously removed from the fuel salt and collected in this OGS mixture where they decay. In the current implementation, these decay-only mixtures have a fixed volume of 1 cm^3. If nuclides are continuously removed into a decay-only mixture, then the concentration of these nuclides continuously increases. For the analysis of the absolute mass of the nuclides, the volume is irrelevant. However, if concentrations (volumetric units) are of interest, then this fixed volume must be appropriately considered.

2.3.2. Specific power

The power for a depletion calculation in TRITON is specified as specific power in unit megawatts per metric ton of initial heavy metal (MTIHM). TRITON calculates the system's initial heavy metal mass $\text{HM}_{\text{initial}}$ at the beginning of the calculation, at $t = 0$, and uses this initial mass as the reference to determine the individual materials' power. If heavy metal is fed to the fuel, then the heavy metal mass—and therefore the total system power—might change. In this case, the specific power provided in the input must be adjusted to consider the additional heavy metal mass HM_{feed} by multiplying the specific power by $1 + \frac{\text{HM}_{\text{feed}}}{\text{HM}_{\text{initial}}}$ and confirming an appropriate step size after which these adjustments are necessary. Future versions of TRITON will permit specifying a total system power when using 3D models to avoid requiring such adjustments.

In depletion calculations of a system in which the fuel is flowing through a reactor core and a loop outside the reactor, but the outside loop is not part of the model, the specific power p must consider the time the fuel spends outside the reactor at zero power (T_{loop}). It is scaled according to the ratio of the resident time in the core (T_{core}) to the whole system as follows:

$$p = \frac{P_{reactor}}{m_{core}} \times \frac{T_{core}}{T_{core} + T_{loop}} \tag{15}$$

However, the resident time of the salt in the core and the loop is not always known. In this, the specific power p can be calculated by dividing the total reactor power P by the total system salt mass (i.e., the sum of the salt mass m inside the core and the loop):

$$p = \frac{P_{reactor}}{m_{core} + m_{loop}} \tag{16}$$

2.3.3. Precursor drift

Precursor drift occurs in systems with flowing fuel where decaying fission products that release neutrons (i.e., delayed neutron precursors) move away from their birth location (i.e., the fission site where they were born). Delayed neutron precursors are critical for reactor safety because they limit the system's dynamic response to reactivity insertions below the effective delayed neutron fraction threshold. This effective delayed neutron fraction ranges from 0.3% to 0.7% of the total neutrons and depends on the fissioning nuclide and other reactor design characteristics. Movement of precursors away from their birth location will decrease this effective delayed neutron fraction because precursors that decay outside of the core, will yield neutrons that do not contribute to the sustained critical condition of the reactor. For MSRs, a reduction in the delayed neutron fraction of up to 50% may be possible.[5]

Many reactor physics tools effectively combine prompt and delayed neutron emission during solutions by using a total fission emission spectrum $\chi(E)$ and total neutrons emitted per fission $\bar{\nu}_t$; delayed neutron emission is often neither explicitly simulated nor tracked. Precursor drift affects this type of solution by (1) changing the number of neutrons emitted per fission and (2) changing the fission emission spectrum because precursors emit neutrons with a softer energy spectrum. Essentially, precursor drift slightly reduces the number of neutrons emitted by fission (less than 0.4%) and slightly increases the emission probabilities of epithermal neutrons from fission. Global steady-state criticality solutions (i.e., those that are used to generate one-group cross sections to perform depletion simulations) are largely unaffected by these small changes. Effects from delayed neutrons drifting into low-power regions are notable, but only in a localized sense. Even in a localized

[5] Experience from the Molten-Salt Reactor Experiment operations has demonstrated robust dynamic characteristics despite the reduction to the effective delayed neutron fraction (Haubenreich and Engel, 1970).

framework, the change in neutron emission spectrum has a negligible effect because the emission of epithermal neutrons is already highly improbable relative to prompt fission emission.

For these reasons, precursor drift has a negligible effect on the longer time-dependent fuel composition in a fluid-fueled system such as an MSR (Betzler et al., 2017a). Its simulation is outside the scope of this article.

3. Implementation and testing

The SCALE implementation for these methods was tested on simple problems representative of potential applications. This includes the point depletion solver ORIGEN and the neutron transport-depletion coupling interface in TRITON. ORIGEN already allows adding a removal constant, but (1) this capability is only accessible directly through the ORIGEN input, and (2) the functionality does not accumulate the amount being removed. The new implementation in TRITON addresses these issues, resolving the dependency graph and solve order, and performing the mapping and integration of removals from a target mixture to sources of a destination mixture.

This functionality in TRITON was demonstrated and verified for a representative three-mixture unit cell problem with an irradiated mixture of ^{233}U (5%) and ^{232}Th (95%) (mixture 1) and initially empty waste mixtures (mixtures 2 and 3). This representation approximates some of the elemental removals in a thorium-fueled MSR system, where the beta decay product ^{233}Pa from neutron capture in ^{232}Th is removed to decay into fissile ^{233}U before recycling back into the core. Protactinium and neodymium are continuously removed from the irradiated mixture and fed into the waste mixtures (Fig. 2); the removal rates are listed in Table 1. In ORIGEN, the feed rates from mixture 1 to mixtures 2 and 3 are obtained such that the amount of protactinium and neodymium in the waste mixtures at the end of time step is consistent between ORIGEN and TRITON. This demonstration used a depletion time step of 10 days in both TRITON and ORIGEN. The optimal time step depends on the removal fraction of the isotope, and a smaller time step is generally recommended to improve the accuracy of the removed isotopes. If necessary, a sensitivity calculation on the time step selection can be performed in advance.

The important isotopes in this three-mixture problem are governed by the following set of equations:

$$\frac{dN^1_{\text{Th-233}}}{dt} = -\lambda_{\text{Th-233}} N^1_{\text{Th-233}}; \tag{17}$$

$$\frac{dN^1_{\text{Pa-233}}}{dt} = -\lambda_{\text{Pa-233}} N^1_{\text{Pa-233}} + \lambda_{\text{Th-233}} N^1_{\text{Th-233}} - \lambda^{1\to2}_{\text{Pa,rem}} N^1_{\text{Pa-233}} - \lambda^{1\to3}_{\text{Pa,rem}} N^1_{\text{Pa-233}}; \tag{18}$$

$$\frac{dN^m_{\text{Pa-233}}}{dt} = -\lambda_{\text{Pa-233}} N^m_{\text{Pa-233}} + \lambda^{1\to m}_{\text{Pa,rem}} N^1_{\text{Pa-233}} \quad \text{for mixtures} \quad m = 2,3; \tag{19}$$

$$\frac{dN^m_{\text{U-233}}}{dt} = \lambda_{\text{Pa-233}} N^m_{\text{Pa-233}} \quad \text{for mixtures} \quad m = 2,3; \tag{20}$$

$$\frac{dN^m_{\text{Nd-148}}}{dt} = \lambda^{1\to m}_{\text{Nd,rem}} N^1_{\text{Nd-148}} \quad \text{for mixtures} \quad m = 2,3; \tag{21}$$

where

- N^m_X is the number density of nuclide X in mixture m,
- λ_X is the decay constant for nuclide X, and
- $\lambda^{m'\to m}_Y$ is the removal constant from mixture m' to mixture m for element Y (Table 1).

The asymptotic behavior of this system of equations yields simple relations that can be used to generate analytical solutions for $N^m_{\text{Pa-233}}$ for $m = 1,2,3$ and $N^m_{\text{U-233}}$ and $N^m_{\text{Nd-148}}$ for $m = 2,3$ (shown as black lines in Figs. 3, 4, and 5), using the $N^1_{\text{Th-233}}$ and $N^1_{\text{Nd-148}}$ concentrations from the ORIGEN calculation as these are defined by the flux in mixture 1.

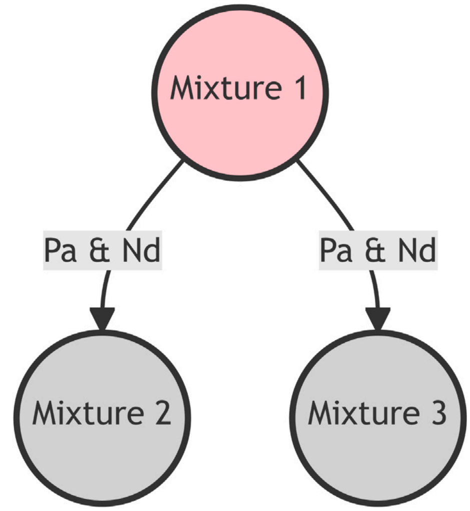

Fig. 2. Diagram of the material flows for three-mixture implementation test.

Table 1
Removal constants for the implementation test problem.

Parameter	Value (1/s)
$\lambda^{1\to2}_{\text{Pa}}$	0.1
$\lambda^{1\to3}_{\text{Pa}}$	0.2
$\lambda^{1\to2}_{\text{Nd}}$	10.0
$\lambda^{1\to3}_{\text{Nd}}$	20.0
$\lambda_{\text{Pa-233}}$	5.29201×10^{-4}
$\lambda_{\text{Th-233}}$	2.97495×10^{-7}

The isotope accumulations in each mixture owing to the constant fractional removal from ORIGEN and TRITON are shown in Figs. 3, 4, and 5 in units of mass per metric ton of initial heavy metal. Because of the relatively high removal rates for protactinium, little protactinium remains in the irradiated mixture during operation (Fig. 3). After about 50 days, the concentration of ^{233}Pa within each waste mixture reaches an equilibrium dictated by its constant generation (removal from mixture 1) and decay rates. The relative ratio of ^{233}Pa in the two waste mixtures is dictated by the magnitude of the removal constants (e.g., twice as much ^{233}Pa is present in mixture 3 relative to mixture 2). The amount of ^{233}U—the decay product of ^{233}Pa (~27 day half-life)—continuously increases because it is not removed and the waste material is not irradiated (Fig. 4). The rate of increase is constant and is proportional to the concentration of ^{233}Pa in the given mixture. Similarly, the stable isotope ^{148}Nd in the waste mixtures continuously increases at a constant rate; the relative ratio in the waste mixtures is again dictated by the removal rate (Fig. 5).

For all investigated masses, TRITON and ORIGEN produce consistent results for $N^1_{\text{Th-233}}$ and $N^1_{\text{Nd-148}}$. The difference between the TRITON sequence and the standalone ORIGEN calculation is that TRITON updates the cross sections and fluxes that are provided to ORIGEN during the depletion simulation (by using a neutron transport calculation—in this case with the discrete-ordinates transport code NEWT), whereas the standalone ORIGEN calculation uses a priori cross section information. Moreover, ORIGEN uses a constant feed rate for mixtures 2 and 3, whereas TRITON applies the fractional removal rate of mixture 1 to build mixtures 2 and 3. Thus, a similar relation as described in Eqs. (18)–(21) still holds (Fig. 2), although an identical result to the ORIGEN simulation will not be obtained via TRITON. The expected asymptotic concentrations of ^{233}Pa in the standalone ORIGEN solution agree with the TRITON-calculated concentrations after about 150 days (Fig. 3). The ^{233}Pa concentrations calculated using TRITON continue to slightly increase in time as the material is irradiated because the fluxes and cross sections are updated with each neutron

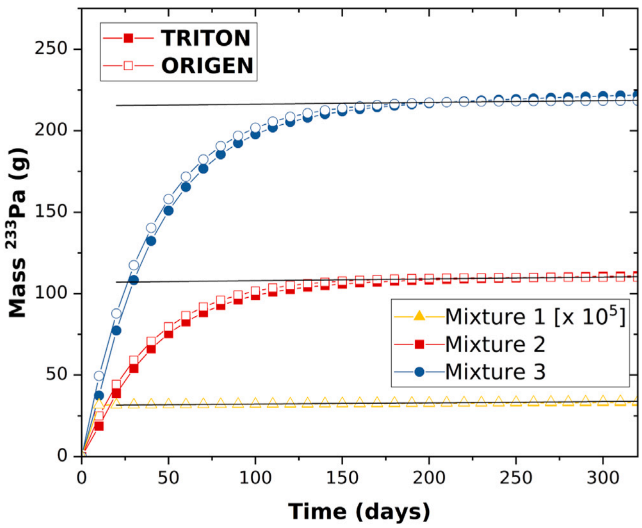

Fig. 3. Mass of ^{233}Pa atoms in the three mixtures in the ORIGEN and TRITON simulations.

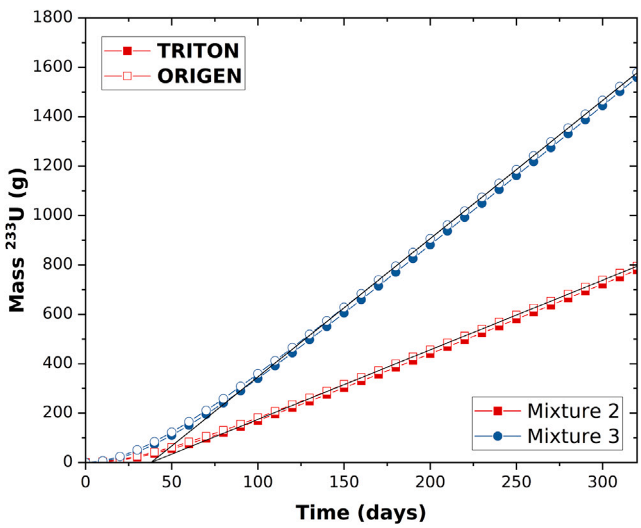

Fig. 4. Mass of ^{233}U in the two waste mixtures in the ORIGEN and TRITON simulations.

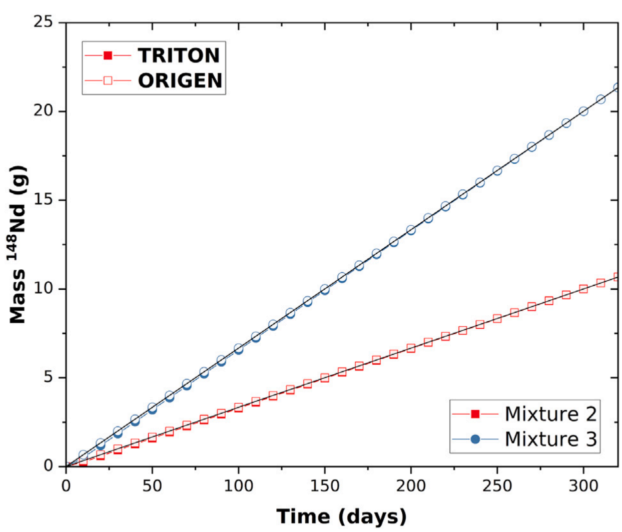

Fig. 5. Mass of ^{148}Nd atoms in the two waste mixtures in the ORIGEN and TRITON simulations.

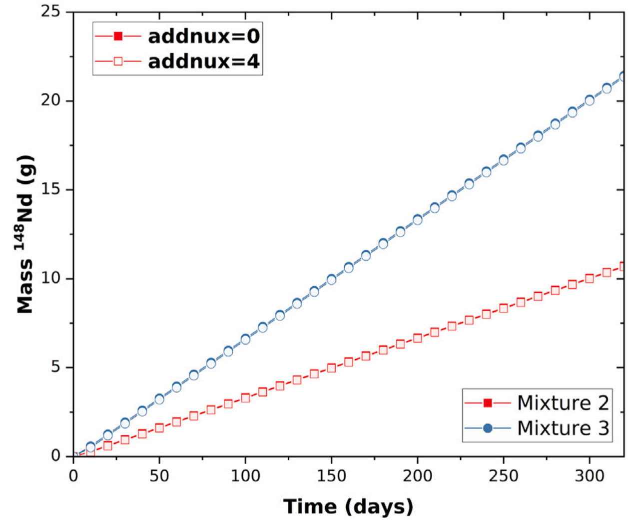

Fig. 6. Mass of ^{148}Nd atoms in the two waste mixtures in TRITON with different *addnux* options.

transport simulation. For both the ^{233}U and ^{148}Nd concentrations in the waste mixtures (2 and 3), TRITON's calculated densities are consistent with ORIGEN, yielding the expected asymptotic rate after a short time (Figs. 4 and 5). These implementation test problems demonstrate that the material flows between mixtures function as expected when using the TRITON input.

Another set of verification tests for the TRITON simulation ensured that the elements removed are properly handed to ORIGEN such that all isotopes of a given element are removed, regardless of whether it has apparent cross-section data. This capability is demonstrated by examining the same TRITON simulation with and without neodymium in the transport calculation. In TRITON, the list of nuclides to be included in the transport calculation is controlled by the *addnux* option: no additional nuclides are considered when *addnux* = 0, and all nuclides for which cross-section data are available in the libraries are included when *addnux* = 4. The ^{148}Nd masses in Fig. 6 agree for the different *addnux* settings, confirming that ORIGEN correctly handles nuclides.

An additional graded set of use cases was designed to identify any potential issues with the solvers or with accurately passing information (Betzler et al., 2019a); however, these solutions are not exhaustively provided herein.

4. Application

The demonstration of TRITON to simulate the flowing fuel was applied to the Molten-Salt Reactor Experiment (MSRE) (Haubenreich and Engel, 1970; Robertson, 1965) for which a SCALE model was previously developed (Bostelmann et al., 2022b; Lo et al., 2022). The MSRE was the first large-scale experiment using a fluid fuel salt and a graphite moderator conducted at the US Department of Energy's Oak Ridge National Laboratory. The reactor reached its first criticality in June 1965 and was designed to generate 10 MWth of power. The MSRE was initially operated using salt containing enriched-^{235}U salt and was later switched to ^{233}U-based salt. The present work focuses on the MSRE fueled with ^{235}U that operated for 9006 equivalent full-power hours (Haubenreich and Engel, 1970).

4.1. Computational model

Performing depletion calculations using a 3D core model is computationally expensive; therefore, a computational model of a 2D axial core slice of MSRE was developed and used in these analyses. The

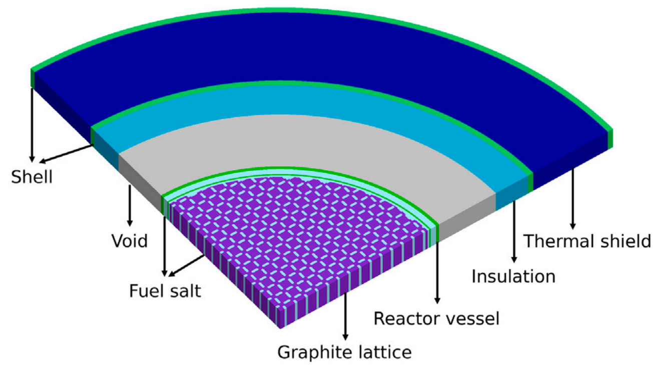

Fig. 7. SCALE model of an axially reflected core slice of the MSRE.

Table 2
MSRE major design parameters (Shen et al., 2019; Fratoni et al., 2020; Shen et al., 2021).

Parameter	Value
Reactor power (MWth)	8
Fuel salt volume (m^3)	1.996
Initial heavy metal mass (MTIHM)	0.218
Specific power (MWth/MTIHM)	36.96
Fuel salt density (g/cm^3)	2.3275
Fuel temperature (K)	911
Fuel salt composition (mol %)	64.88 LiF
	29.27 BeF_2
	5.06 ZrF_4
	0.79 UF_4
Graphite density (g/cm^3)	1.8507
Graphite temperature (K)	911
Graphite lattice radius (cm)	780.285
Inner core can radius (cm)	71.097
Outer core can radius (cm)	71.737
Inner reactor vessel radius (cm)	74.299
Outer reactor vessel (inactive region) radius (cm)	75.741
Individual stringer width (cm)	5.084
Fuel channel thickness (cm)	1.018
Fuel channel width (cm)	3.053
Cutout radius at fuel channel corners (mm)	0.508

model was developed based on the MSRE benchmark compiled in the International Reactor Physics Experiment Evaluation Project (IRPhEP) handbook (Shen et al., 2019; Fratoni et al., 2020; Shen et al., 2021). Fig. 7 depicts the quarter view of the TRITON model, and Table 2 lists key parameters used in modeling. The MSRE core comprised a lattice of rectangular prism-shaped graphite “stringers” that were vertically oriented within a cylindrical reactor vessel through which the salt was pumped upward, flowing through the channels between the stringers. The slice model developed for this work is a slice through the core’s axial midplane; vacuum boundary conditions are applied radially, and reflective boundary conditions are applied axially (i.e., it simulates an axially infinite height). Using an axial core slice model is part of the modeling strategy for the one-group cross-section (ORIGEN library) generation for other graphite-moderated non-light-water reactors for fuel inventory generation (Skutnik and Wieselquist, 2021; Bostelmann et al., 2022a).

To demonstrate that a slice model provides an adequate surrogate model for depletion compared with a full-core model, the neutron flux in both models was compared. Fig. 8 shows that the neutron flux in the center of the slice model agrees very well with the corresponding neutron flux in the center of the 3D full-core model. Although using a small unit cell was insufficient, a model with a similar moderator-to-fuel ratio as the full core and with radial leakage adequately represents the full core in terms of neutron flux and therefore the fuel salt inventory during irradiation.

In this work, TRITON was used to deplete the slice model for 375 days, corresponding to the MSRE’s total operation time with ^{235}U.

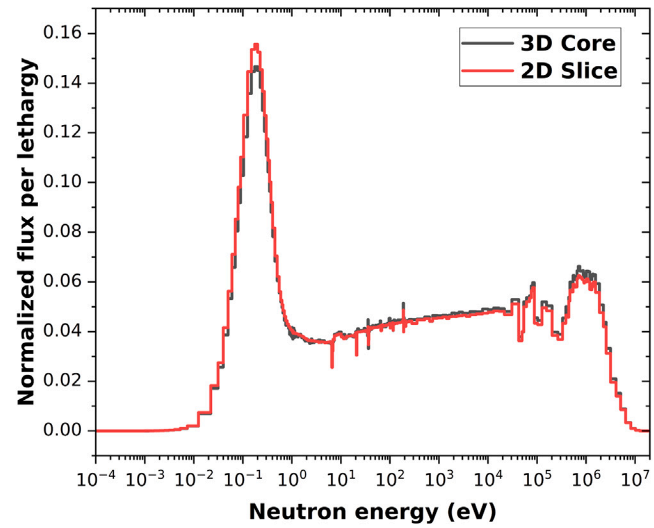

Fig. 8. Normalized neutron flux in the center of the slice model and the center of the full-core model.

TRITON applied the Monte Carlo code KENO-VI for the neutron transport calculation to update the flux and cross sections. KENO-VI was applied in continuous-energy mode using ENDF/B-VII.1 nuclear data libraries (Chadwick et al., 2011). Each neutron transport calculation was converged to reach a statistical uncertainty of the eigenvalue of 20 pcm. To provide a code-to-code comparison, the TRITON results were compared with corresponding Serpent (Leppänen et al., 2015) results. An identical Serpent slice model was developed and simulated with the same calculation conditions as with TRITON, including the application of Serpent’s online removal and continuous feeding features with consistent settings (Aufiero et al., 2013).

4.2. Removal rates

In this calculation, two classes of elements were continuously removed during the operation: noble gases and noble metals, as illustrated in Fig. 9. The noble gases such as krypton and xenon were actively removed from the fuel salt via an OGS. They were held in a holding tank before entering the charcoal bed where they remained for several days before being filtered, monitored, and then released into the environment. For simplicity, the holding tank and filter system are not shown in Fig. 9. The noble metals were reduced in the fuel salt by depositing (i.e., plating out) on the surfaces within the loop (e.g., heat exchanger, pipes, graphite surfaces in-core, OGS). Kedl (1972) reported that 99% of the noble metal fission products resided outside the core, and some of them decayed later to halogens iodine and bromine.

To obtain the accurate isotopic compositions of the fuel salt, the removal rates for calculation model (Table 3) were derived from the experimental data, as follows:

1. The removal rate of noble gases from the fuel salt to the OGS was obtained by considering the xenon poison fraction. This quantity is defined as the ratio of macroscopic absorption of ^{135}Xe to ^{235}U, which varies between about 0.3% and 0.4% (Kedl and Houtzeel, 1967).
2. The removal rate from the OGS to the charcoal bed was derived by considering the hold time at the piping (1.5 h) to allow the short-lived fission products to decay, further decreasing the heat generated in the charcoal bed (Robertson, 1965).
3. The removal rate of noble gases from the charcoal bed to the vent considered the hold time (90 days for xenon and 7.5 days for krypton) to allow the gaseous fission products to decay to stable elements (Robertson, 1965).

Table 3
Nuclide removal rates applied in the MSRE calculation.

From	To	Class	Elements	Rates (s^{-1})
Fuel salt	OGS	Noble gases	Xe, Kr	4.067×10^{-5}
OGS	Charcoal bed	Noble gases	Xe, Kr	1.388×10^{-4}
Charcoal bed	Vent	Noble gases	Xe, Kr	8.914×10^{-8}
Fuel salt	Structural surfaces	Noble metals	Se, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Sb, Te	8.667×10^{-3}
Structural surfaces	Fuel salt	Halogen	I, Br	1.000×10^{0}

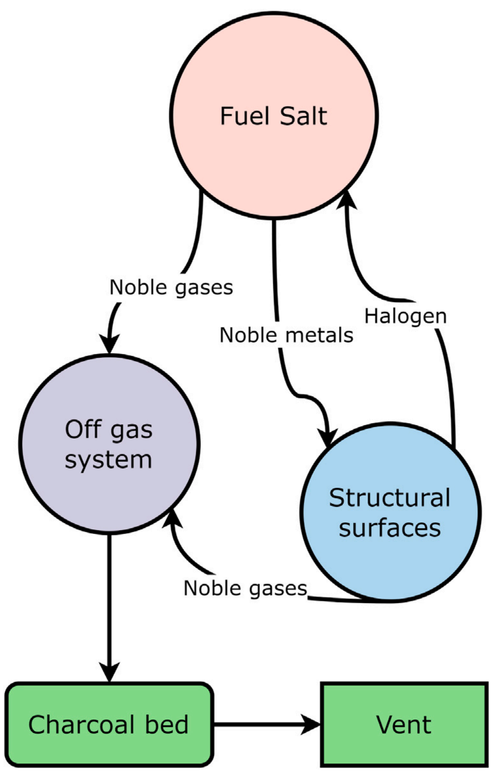

Fig. 9. Diagram showing material removals considered in the MSRE calculation.

4. The removal rate of noble metals from the fuel salt by plating out was calculated by considering the mass transfer rate and the surface area of the fuel salt loop component (Kedl, 1972).
5. Iodine and bromine produced by the decay of some of the noble metals were returned to the primary fuel salt.

4.3. Removed materials

The effect of removing xenon and krypton to the OGS on the xenon and krypton densities in the fuel salt is shown in Fig. 10. In a regular depletion calculation without removal, xenon and krypton densities increase over time because fission and decay products accumulate; removing xenon and krypton from the OGS causes lower densities that converge to approximately constant values after a few days. The removed amounts of xenon and krypton at the end of the depletion calculation are approximately 30.6 L. TRITON and Serpent demonstrate excellent agreement in terms of estimating the inventory of xenon and krypton in the core and in the OGS.

The effect of the xenon removal on reactivity is illustrated in Fig. 11. Because removing xenon results in less neutron absorption from ^{135}Xe, the eigenvalue k of the slice is between 750 and 930 pcm higher compared with a system from which xenon is not removed. The eigenvalue at the beginning of a cycle is consistent between TRITON and Serpent: the relative difference is about 19 pcm. The initial small difference increases throughout depletion as an aggregate result of

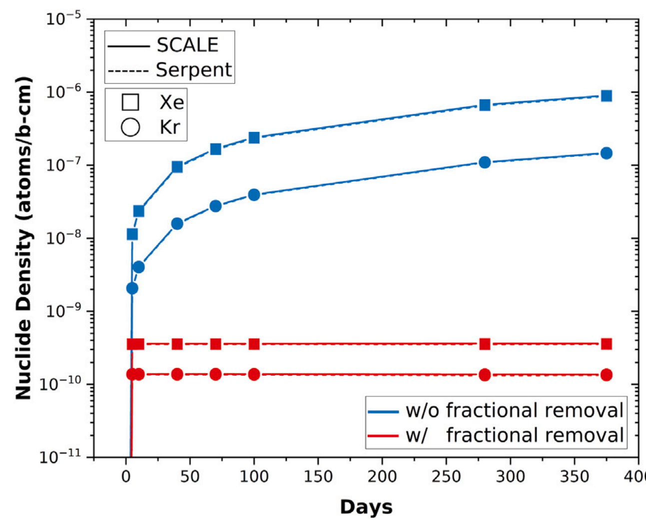

Fig. 10. Xenon and krypton densities in the fuel salt with and without considering their removal.

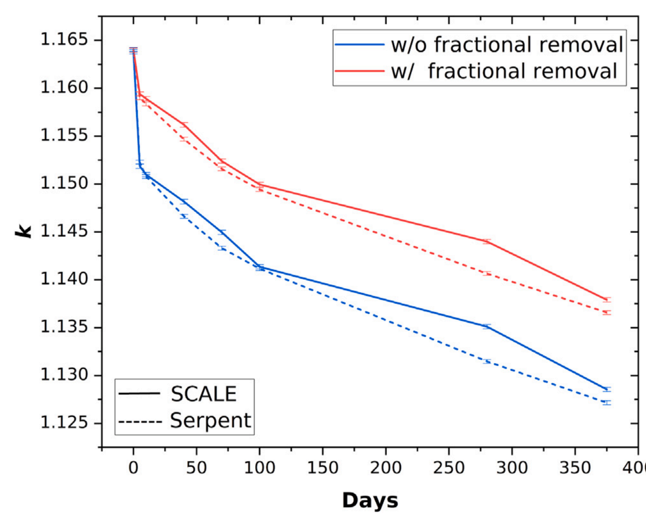

Fig. 11. System eigenvalue k of core slice with and without considering xenon and krypton removal.

various differences in code methodologies and associated nuclear data such as energy release from fission and capture, power normalization, depletion algorithms, and applied data processing between SCALE and Serpent. This result has been observed elsewhere (Bostelmann et al., 2020). However, the large discrepancy in k is not reflected in the isotopic inventory between the two codes, as discussed previously and subsequently.

The isotopic composition at each location can be inspected individually because it is tracked as an individual mixture. The ability to track and decay removed materials accurately is the additional functionality required from the simulation to determine storage needs, characterize source terms, and define material feedback rates. For example, the

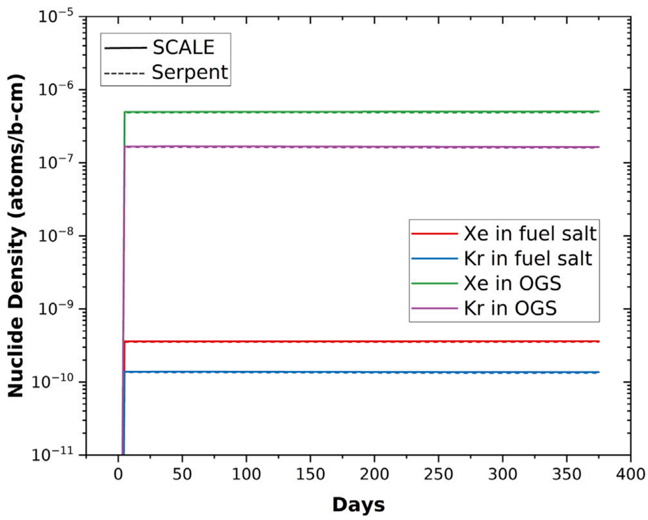

Fig. 12. Densities of xenon and krypton in the fuel salt and the OGS.

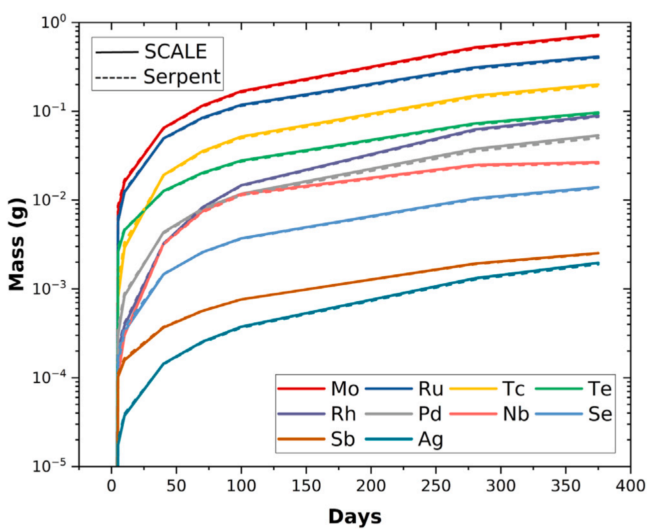

Fig. 13. Densities of plated-out noble metals.

xenon and krypton densities in the salt and OGS are shown in Fig. 12; they are constant because they are constantly removed. The accumulation of the plated-out noble metals is depicted in Fig. 13. The results also demonstrate that TRITON and SCALE provide consistent results.

4.4. Continuous ^{235}U feed

The new feed functionality in TRITON is demonstrated by an additional MSRE core slice depletion case in which the fissile isotope ^{235}U is continuously added. The feed rate is required by TRITON in units of mass or mole per time per metric ton of initial heavy metal. The amount of the fed ^{235}U is estimated based on the total consumed ^{235}U mass during the depletion without any feed. Because the consumed ^{235}U mass was about 17.643 kg/MTIHM after 375 days of operation, the feed rate of ^{235}U was determined to be 5.445×10^{-4} g/(MTIHM s). The constant mass of ^{235}U over time is illustrated in Fig. 14. The figure also shows the consistent results between SCALE and Serpent.

5. Conclusion

The capability to simulate irradiation with continuous material feeds and removals is implemented in the SCALE 6.3 code system

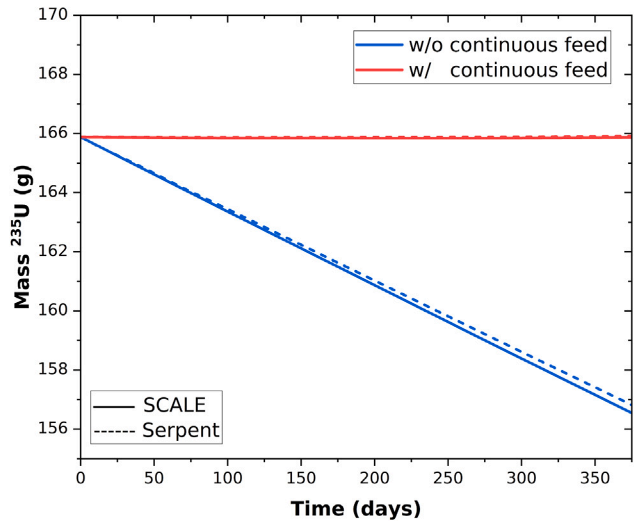

Fig. 14. Mass of ^{235}U of core slice with and without continuous feed.

for predicting the isotopic composition throughout months to years of operation in advanced liquid-fueled nuclear reactor systems, including MSRs. The ORIGEN depletion and decay solver and the TRITON reactor physics sequence were extended to accept material feed definitions and to track and decay removed materials, enabling the long-time simulation of the complex material flow paths characteristic of liquid fuel systems. Capabilities are accessible via new input definitions in TRITON. The implementations were tested and demonstrated for simple representative MSR use cases—including an example case of MSRE.

Integrating these capabilities within the SCALE code system provides reactor developers and analysts access to a quality-assured tool set to reliably inform fuel-cycle and reactor system design for liquid-fueled systems. The flexibility within the tool set allows for application to a variety of potential reactor designs, appropriate for the ever-evolving advanced nuclear reactor landscape. The breadth of available modules within SCALE provides for the extension to other relevant capabilities, including sensitivity and uncertainty analyses, optimization approaches, and decay heat assessments. The accurate predictions of time-dependent liquid fuel isotopic composition throughout months to years of reactor operation are critical for reactor and processing system design, source term analyses, and safeguards approach design. Ultimately, these tool sets provide the time-dependent isotopic information critical to informing the design and deployment of advanced liquid-fueled reactor systems.

CRediT authorship contribution statement

Donny Hartanto: Methodology, Formal analysis, Data curation, Writing – original draft, Visualization. **Friederike Bostelmann:** Methodology, Software, Formal analysis, Data curation, Writing – original draft, Project administration. **Benjamin R. Betzler:** Conceptualization, Methodology, Software, Writing – original draft. **Kursat B. Bekar:** Software, Validation, Writing – review & editing. **Shane W. Hart:** Software, Validation, Writing – review & editing. **William A. Wieselquist:** Conceptualization, Methodology, Software, Writing – original draft, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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